TrailTrack

Real-Time Trail Conditions and Community Walks for Ireland

Iteration 1 Report: Release v0.1

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1. Summary

Iteration 1 built the foundations of TrailTrack. Trail routes are imported from the Sport Ireland National Trails Register and shown on a map. Users can create an account, report a trail's condition, and search for trails by name or by distance. All of this is in release v0.1, for the 28 trails in County Laois.

Every feature in the specification's plan for the iteration was delivered, along with three from the wireframes: a trail detail screen, a button to centre the map on the user, and bottom navigation. The work was tracked as 14 issues and merged through 8 pull requests. Table 1 lists what the release contains.

Feature	What v0.1 does
Trail import	An administrator imports the Laois trails from the register. Running it again updates them instead of creating duplicates.
Map	Routes in the colour of their trail signs, grouped into sites, with a chooser where routes share a path.
Accounts	Registration and login with an email and password. Logins last 7 days and survive the app being closed.
Trail details	Facts, description, current conditions and the reports from the last 7 days.
Condition reports	Logged-in users report one of 9 condition types with a severity and an optional note. Anyone can read reports.
Search	By name, or near the user's position within 5, 10 or 25 km, with each trail's latest condition.
Navigation	Map, Search and Profile tabs. A button centres the map on the user.

Table 1: Features in release v0.1

2. What Worked

- **Importing the register once.** Trails are imported into PostgreSQL by an administrator, not fetched from the register on every request. The app keeps working if the register is slow or changes its format.
- **PostGIS for the spatial work.** It simplifies each route for the map (about 78 KB for all 28 trails), groups trailheads into sites, and finds trails within a distance using a spatial index.
- **Integration tests against the real database.** They found bugs that unit tests with mocks could not. For example, the reference that stops a report being stored twice was never saved, and only a test against PostgreSQL showed it.
- **Checking each feature on the emulator.** Running against the real backend found layout bugs that tests did not, such as the map fitting to the whole world and the system gesture bar covering the map credits.
- **Planning in GitHub.** Each release has a milestone, each feature an issue, and each feature a branch and pull request. The board shows what is left at a glance (Figure 3).
- **MVVM, introduced gradually.** New screens keep their state in a view model, so their logic is tested with fake services instead of a running backend.

3. What Did Not Work

- **A temporary account button.** The login screens were built before the bottom navigation shown in the wireframes. A small account button on the map stood in for it, then was removed when the navigation was built.
- **The database schema is generated.** Hibernate creates the tables, but it cannot create the spatial index the distance search needs, so a class creates that index at startup. Database migrations are planned for the start of Iteration 2.
- **Location on the emulator.** The usual command for setting the emulator's position did not change Android's location, so "Near me" could only be tested once a test location provider was used.

- **Documentation tools.** Doxygen does not support Dart, so the app is documented with dart doc. Doxygen also skips Java records, so a small filter presents them to it as classes.
- **The map screen is not MVVM yet.** It was built before the view models and still holds its own state.

4. Problems and Solutions

Table 2 lists the main problems found during the iteration and how each was solved.

Problem	Cause	Solution
Cross-county trails were missing.	The County field lists every county a trail crosses, e.g. "Laois, Offaly". An exact match found 24 of the 28 trails and dropped the Slieve Bloom Way and the Barrow Way.	Filter with County LIKE '%Laois%'.
Coordinates were not latitude and longitude.	The register returns Irish Grid coordinates (EPSG:29903) by default.	Request WGS84 (outSR=4326), which PostGIS and the map library expect.
Routes had no start point.	The register's Latitude and Longitude fields are empty for every Laois route.	Take the official start from the register's trailheads layer, matched by trail ID.
The map zoomed out to the whole world.	The camera was fitted to the trails before the map had a size.	Fit the camera only once the map reports that it is ready.
Routes from one car park overlapped.	Several loops start at the same trailhead and share paths.	Group trailheads within 10 m of each other into sites with PostGIS (28 routes in 13 sites). Each site has a pin, and chips switch between its routes.
Shared paths hid each other.	Routes on the same path were drawn on top of each other.	Draw longer routes first and wider, so shared paths show nested colour bands. A tap on a shared path asks which route was meant.
A report could be stored twice.	A report sent again after a dropped connection looked like a new one.	The app gives each report a unique reference. The server returns the stored report when it sees the same reference again.
Google asked to collect location data.	The location plugin used Google Play services by default, which shows a data-sharing prompt.	Use Android's own location manager instead.
"Near me" sometimes searched by name only.	Choosing a radius before the position was found ran a search without it.	Wait for the position before searching. Found while recording the screencast.

Table 2: Problems found in Iteration 1 and their solutions

5. From Difficulty Colours to Waymark Colours

The first version of the map coloured each route by difficulty: easy routes in solid green, moderate in dashed orange and challenging in dotted red. It looked clear, but it did not match what walkers see on the ground.

Official trail maps, such as Coillte's maps for Oughaval Woods, Glenbarrow and Monicknew, identify each route by the colour of its waymark arrows, never by difficulty. Where routes share a path, they draw the coloured lines side by side. A walker at a trailhead follows a colour, so the app should use the same colours.

The register describes each route's signs in a text field, for example "Blue arrow on white background". The import turns this into a colour: the arrow's colour, or the background's when the arrow is white. A yellow arrow on black marks a National Waymarked Way, drawn in yellow with a black edge. Four routes have no usable colour and get neutral colours instead. Difficulty and length moved into the route card (Figure 1).

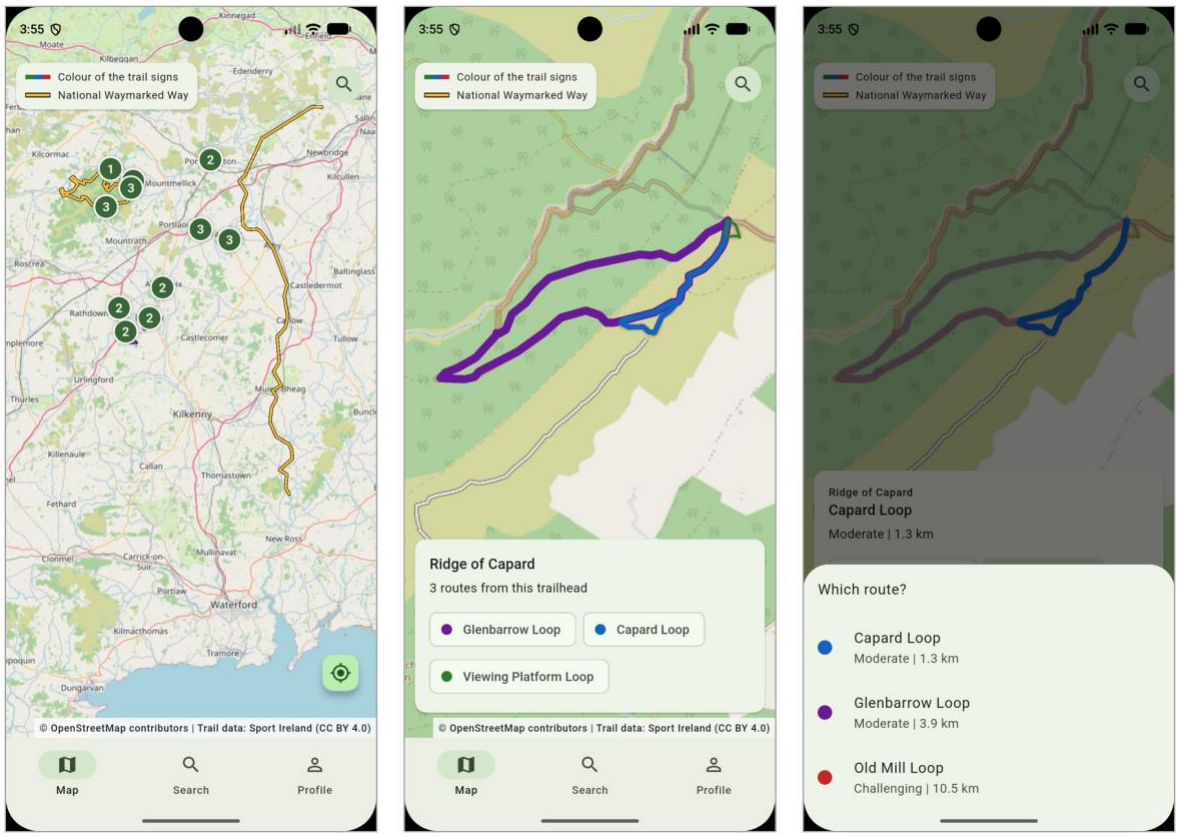


Figure 1: The map, a site with its routes, and the route chooser on a shared path

6. How the Register's Fields Were Found

The register is published as two ArcGIS feature services: trail routes and trailheads. Neither has written documentation of its fields, so they were found in two steps.

- **Layer metadata.** Adding ?f=json to a layer's address returns its field names and types, its geometry type and its coordinate system. This showed the fields to import and that coordinates default to the Irish Grid.
- **Test queries in Postman.** Each assumption was checked before the import relied on it. Counting with returnCountOnly compared the exact and LIKE county filters (24 against 28). Querying fields without geometry showed that Latitude and Longitude are empty. Listing the trailhead types showed Start, Finish and Secondary, of which only Start marks a route's official start.

The requests are saved as a Postman collection, so the checks can be repeated if the register changes. The same approach ruled out another data source: Coillte's felling licence layer is an archive of applications up to 2025, not a live list of closures.

7. Testing

Part	Tests	What they cover
Backend	70	Unit tests of services, web layer tests including the 401 and 403 security responses, and integration tests against PostgreSQL with PostGIS, rolled back after each test.
App	101	Unit tests of models, API clients and view models with fake services, and widget tests of each screen on a phone-sized display.

Table 3: Automated tests in v0.1

Each feature was also checked on the Android emulator against the real backend before its pull request was merged. Figure 2 shows search, the report form, and a new report on its trail.

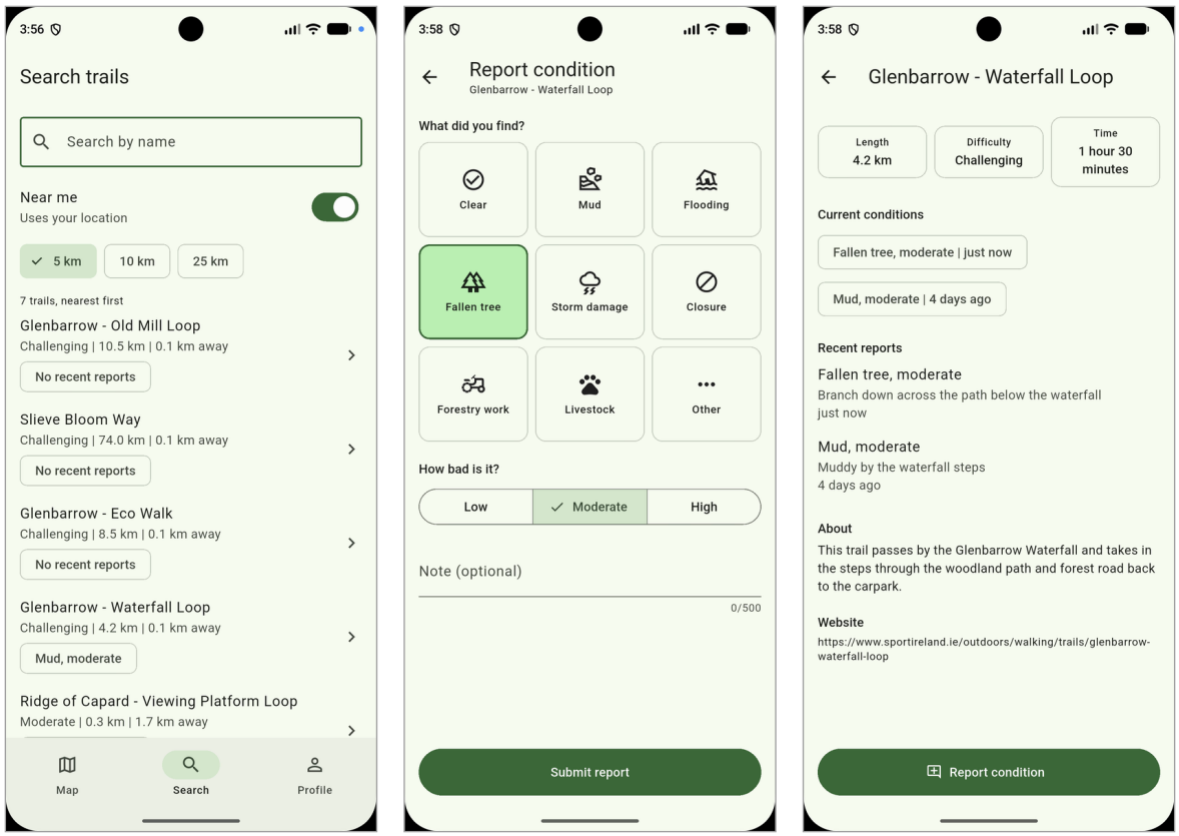


Figure 2: "Near me" search, the report form, and the trail with the new report

8. Plan for Iteration 2

Iteration 2 runs to release v0.2, due on 4 December. Its main feature is community walks, the project's main difference from existing general-purpose hiking platforms. Table 4 lists the planned work; each item will be broken into smaller issues at the start of the iteration.

Issue	Work
#21	Database migrations with Flyway, replacing the generated schema and the startup index. Refine the data model, including when reports expire.
#16	Community walks: create a walk on a trail, with a date, meeting point and attendee limit.
#17	Community walks: browse upcoming walks and RSVP. Attendee lists are visible only to attendees and the organiser.
#18	Live condition overlays on the map, so a trail's current state shows without opening it.
#19	An optional hazard location on reports, snapped to the trail and shown as a pin, never with the reporter.
#20	Filter trail search by difficulty.

Table 4: Planned work for Iteration 2

Before building the walks, I will show the idea to a local walking club to check that it fits how they organise walks today. If time allows, the map screen will move to a view model like the other screens.

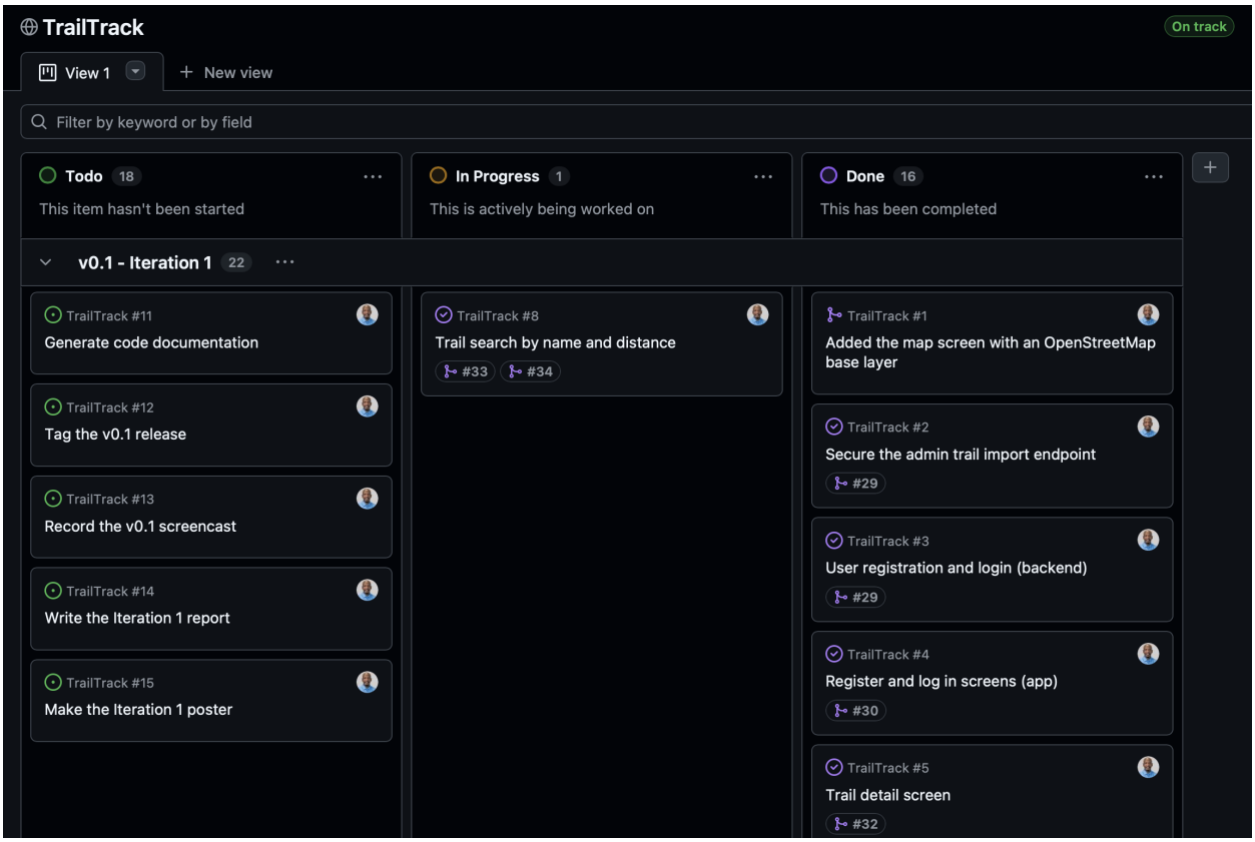


Figure 3: The TrailTrack project board